Do Now · retrieval

Noughts & Crosses · Part 3 — Make moves

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which method from Part 2 displays the board?

Q2

In a zero-indexed C# 2D array, which row and column does board[1,2] access?

Q3

What do nested loops let us do with the board?

Q4

Move 5 is the centre square. What zero-based row and column should it map to?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Map 1–9 into row/column — CODE:
int row = (move - 1) / 3;
int col = (move - 1) % 3; — Integer division finds the row; MOD finds the column. The picture highlights move 8: row 2, column 1. Inside MakeMove(), check the 1–9 range before using these indexes.

Core idea 2

Check the move before accessing the array — CODE:

```
if (move < 1 || move > 9)
return false;

// Now calculate row and col.
// Only access the array for a valid index.
if (board[row,col] == "X" ||
board[row,col] == "O")
{
return false;
```

} — Reject choices outside 1–9 before calculating/using their indexes. Reject occupied squares too. Never overwrite an X or O.

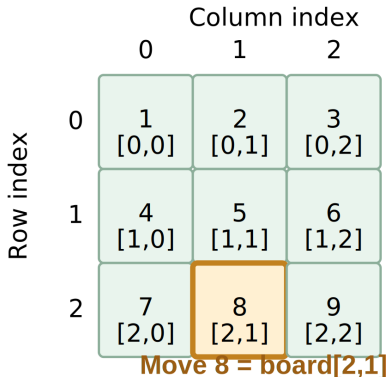
Core idea 3

Make the move — CODE:

```
board[row,col] = player;
return true;
```

— A Boolean return value tells Main whether the move succeeded.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Switch player only after success — CODE:
if (currentPlayer == "X")
currentPlayer = "O";
else
currentPlayer = "X"; — An invalid move should leave the same player to try again.

Vocabulary · use the correct technical terms

2D array — An array organised using rows and columns.

Row — The horizontal index in a 2D array.

Column — The vertical index in a 2D array.

Nested loop — A loop inside another loop.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Turn move 8 into a board position

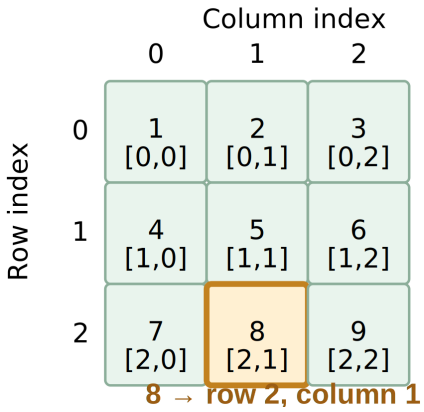
Follow the worked method

1. Subtract 1 — CODE:
move - 1 = 7 — Board indexes begin at 0.

2. Use MOD 3 — CODE:
7 % 3 = 1
col = 1 — The remainder identifies the column.

3. Place the player — CODE:
board[2,1] = player; — Move 8 maps to row 2, column 1.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the same calculation
CODE:
row = (6 - 1) / 3 = ?
col = (6 - 1) % 3 = ?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Map a move

Write code that converts an integer move from 1–9 into zero-based row and column values.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Make alternating turns work
Use your Part 2 DrawBoard() , prompt for a move, call MakeMove() , and switch player only after a valid move.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

If a player chooses an occupied square, the program should switch to the other player anyway.

Use key vocabulary from memory

1. Explain 2D array in your own words.

2. Explain Row in your own words.

3. Explain Column in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.